

GEOLOGICAL FEATURES

Thanks to space-probe exploration, astronomers have a full and detailed picture of Venus's varied landscape. The planet has three main highland regions, termed terra. They are Aphrodite, which dominates the equatorial zone, and Lada and Ishtar. Over 20 smaller, upland areas, termed regio, are found around the planet. Extensive lowland plains, termed planitia, complete the global picture. Volcanic activity is evident across most of the surface, but the volcanoes are not randomly distributed. There are more in the uplands, particularly in Atla and Beta Regio, than in the highlands or plains.

VOLCANIC TERRAIN

This view across western Eistla Regio is typical of the Venusian surface. The volcanoes on the skyline are Sif Mons (left) and Gula Mons.

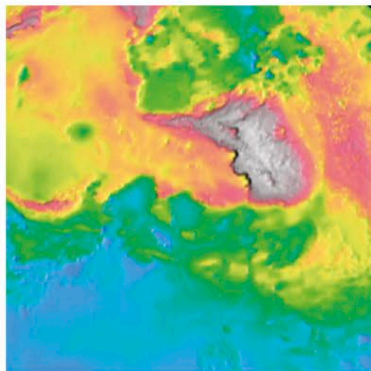
ISHTAR TERRA

Ishtar Terra



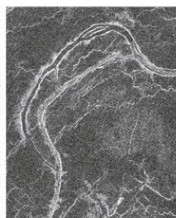
TYPE Highland terrain
AGE Under 500 million years
LENGTH 3,485 miles (5,610 km)

Ishtar Terra is a large plateau about the size of Australia, which stands 2 miles (3.3 km) above the surrounding lowlands. It is the nearest thing on Venus to the continents on



LAVA CHANNEL

Running for well over 1,200 miles (2,000 km), this lava channel is unusually long.



Earth. Its western region is the elevated plateau Lakshmi Planum, which is bounded at the northwest by the Akna Montes and the Freyja Montes and to the south by the Danu Montes. The steep-sided Maxwell Montes range makes up the eastern part of Ishtar Terra, along with a deformed area, Fortuna Tessera, to the mountain range's north and east. The plateau was possibly formed as areas of planetary crust were driven together. It is likely that beneath Ishtar there is cooled, thickened crust that is kept up by a rising region of mantle.

MASSIVE PLATEAU

This map, color-coded for height as mapped by Magellan, includes the highest point on the planet, Maxwell Montes. Blue represents the lowest elevation and white is the highest.

ISHTAR TERRA

Akna Montes



TYPE Mountain range
AGE Under 500 million years
LENGTH 515 miles (830 km)

Forming the western border of Lakshmi Planum, Akna Montes is a ridge belt that appears to be the result of folding due to northwest-southeast compression. The mountain building is thought to have occurred after the plains formed, as the plains in this region seem to be deformed.



FOLDING DUE TO COMPRESSION

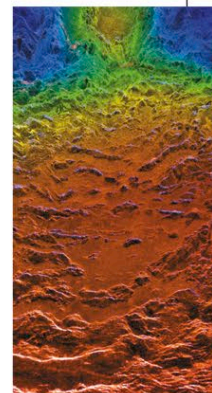
ISHTAR TERRA

Fortuna Tessera



TYPE Ridged terrain
AGE Under 500 million years
LENGTH 1,739 miles (2,801 km)

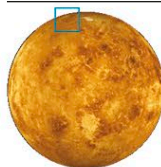
Fortuna Tessera is an area of north-south-trending ridges about 600 miles (1,000 km) wide. The distinctive pattern made by the region's intersecting ridges and grooves led to this type of terrain originally being called parquet terrain, after its resemblance to woodblock flooring, although it is now termed tessera. The image here is a view looking westward across about 155 miles (250 km) of Fortuna Tessera toward the slopes of Maxwell Montes (colored in blue).



RIDGES

ISHTAR TERRA

Lakshmi Planum



TYPE Volcanic plain
AGE Under 500 million years
LENGTH 1,456 miles (2,345 km)

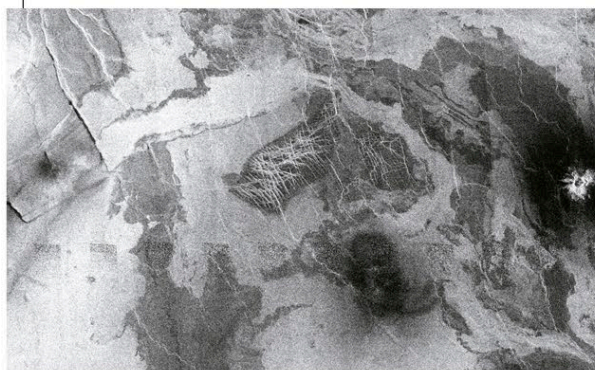
The western part of Ishtar Terra consists of Lakshmi Planum. This is a smooth plateau 2.5 miles (4 km) high, formed by extensive volcanic

eruptions. The plateau is encircled by curving mountain belts—the Danu, Akna, Freyja, and Maxwell Montes—and steep escarpments such as Vesta Rupes to its southwest. This massive plain covers an area that is about twice the size of Earth's Tibetan Plateau (see pp.132–133). Two large volcanic features, the Colette Patera and Sacajawea Patera (see opposite), which punctuate the otherwise relatively smooth plain, were identified in Venera 15 and 16 data. Their floors lie over 1.5 miles (2.5 km) below the plateau level. There are just a few planums on

Venus, all named after goddesses. Lakshmi is the Indian goddess of love and war.

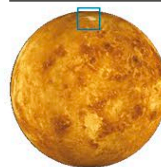
LAVA FLOWS

The eastern Lakshmi region is covered by lava flows. The dark flows are smooth and the light ones are rough in texture. A bright impact crater can be seen on the right.



ISHTAR TERRA

Maxwell Montes



TYPE Mountain range
AGE Under 500 million years
LENGTH 495 miles (797 km)

The Maxwell Montes mountain range forms the eastern boundary of Lakshmi Planum. It is the highest point on Venus, rising over 6 miles (10 km) above the surrounding lowlands. In

its higher regions, the ridges, which are 6–12 miles (10–20 km) apart, have a sawtooth pattern. The mountains fall away to Fortuna Tessera to the east. The western side is a complex of grooves and ridges and is particularly steep—Magellan data revealed that the southwestern flank has a slope of 35°. The mountain range was formed by compression and crustal foreshortening, which produced folding and thrust faulting. Venusian mountain ranges are usually named after goddesses, but Maxwell Montes is named after the British physicist James Clerk Maxwell, a pioneer of electromagnetic radiation.

STEEP SLOPES

This computer-generated image, looking eastward toward the Maxwell Montes, has been colored to show the presence of iron oxides on the surface.

